

a two-class prediction problem, the last stage of the MLP classifier is a two-unit softmax, the output of which gives the probabilities for each of the two output classes. This is shown in the overall block diagram in Figure 1.

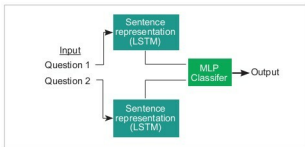


Figure 1: Block diagram for duplicate question detection system

Given that we discussed the overall structure of our implementation, I request our readers to implement this using a deep learning library of their choice. I would recommend using Tensorflow, PyTorch or Keras. We will discuss the Tensorflow code for this problem in next month's column. Here are a few questions for our readers to consider in their implementation:

- How would you handle 'out of vocabulary' words in the test data? Basically, if there are words which do not have embeddings in either Word2vec/Glove or even in the case of corpus-trained embedding, how would you represent them?
- Given that question sentences can be of different lengths, how would you handle the variable length sentences?
- On what basis would you decide how many hidden layers should be present in the MLP classifier and the number of hidden units in each layer?

I suggest that our readers (specifically those who have just started exploring ML and NLP) can try implementing the solution and share the results in a Python jupyter notebook. Please do send me the pointer to your notebook and we can discuss it later in this column.

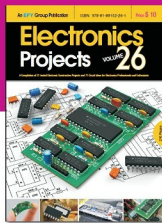
If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers a very happy and prosperous new year! 🐧

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The author is an expert in systems software and is currently working as a research scientist at Conduent Labs India (formerly Xerox India Research Centre). Her interests include compilers, programming languages, file systems and natural language processing. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training (India)' at <http://www.linkedin.com/groups?home=&gid=2339182>.

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